

# Wenbo Lyu

B.Sc. Graduate

## Curriculum Vitae

Saturday 13<sup>th</sup> June, 2026

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## Some stuff about me

- My research interests lie in **advancing methodologies in spatial causal inference** and **developing high-performance computational tools**, with a primary focus on *R packages*.
- I specialize in *data analysis*, *statistical modeling*, and developing open-source analytical tools, including *R packages*, using **R**, **C++**, and **Python**, with a strong focus on *spatial analysis*. I actively contribute to the R geospatial community and am dedicated to advancing open-source geospatial software.
- Currently, my work centers on **Empirical Dynamic Modeling (EDM)** framework for modeling *dynamic system*, **information theory** for quantifying *information flow* and *variable interactions*, **ordinary differential equations (ODEs)** for characterizing *spatiotemporal processes* and *system evolution*, as well as **counterfactual** and **potential outcomes** framework for estimating *causal effects*. I am particularly interested in leveraging these approaches to address critical challenges in *urban sustainability*, *climate change mitigation*, and broader global issues.

## Education

2021.8-2025.6 **B.Sc. In Geographic Information Science**  
Xi'an, Shaanxi

Shaanxi Normal University

## Research Experience

2026.6-2026.11 **Research Assistant**  
Hong Kong

The Hong Kong Polytechnic University

2024.8-2026.6 **Research Intern**  
Guangzhou, Guangdong

The Hong Kong University of Science and Technology (Guangzhou)

## Publications

1. Lyu, W., Dai, S., Song, Y., Zhao, W., Yi, W., Xiao, Y., & Jia, N. (2026). Measuring causal strengths from spatial cross-sectional data with geographical cross mapping cardinality. *International Journal of Geographical Information Science*, 127. <https://doi.org/10.1080/13658816.2026.2687121>
2. Lyu, W., Lei, Y., Yi, W., Song, Y., Li, X., Dai, S., Qin, Y., & Zhao, W. (2026). Causal discovery in urban data with temporal empirical dynamic modeling: The R package tEDM. *Computers, Environment and Urban Systems*, 127, 102435. <https://doi.org/10.1016/j.compenvurbsys.2026.102435>
3. Lv, W., Lei, Y., Liu, F., Yan, J., Song, Y., & Zhao, W. (2025). gdverse: An R package for spatial stratified heterogeneity family. *Transactions in GIS*, 29(2), e70032. <https://doi.org/10.1111/tgis.70032>
4. Lv, W., Liu, F., Cai, K., Cao, Y., Deng, M., Liang, W., Yan, J., & Wang, G. (2024). Distinguishing the impacts and gradient effects of climate change and human activities on vegetation cover in the weihe river basin, china. *Journal of Geophysical Research: Biogeosciences*, 129(10), e2024JG008297. <https://doi.org/10.1029/2024jg008297>
5. Chen, C., Song, Y., Lv, W., Shemery, A., Hampson, K., Yi, W., Zhong, Y., & Wu, P. (2025). Predicting pavement cracking performance using laser scanning and geocomplexity-enhanced machine learning. *Computer-Aided Civil and Infrastructure Engineering*. <https://doi.org/10.1111/mice.13489>
6. Xiao, Y., Yang, Y., Zhao, T., Wang, W., Lv, W., & Zhao, W. (2026). Contrasting causal pathways of vegetation greening between economically strong and weak towns in china's greater bay area. *GIScience & Remote Sensing*, 63(1), 2623327. <https://doi.org/10.1080/15481603.2026.2623327>

Note: Publications under "Lyu, W." and "Lv, W." both refer to Wenbo Lyu.

## Unpublished

|                       |                                                                                                                  |                                            |
|-----------------------|------------------------------------------------------------------------------------------------------------------|--------------------------------------------|
| Co-First Author (1st) | <b>Quantifying pairwise directional influence between spatial variables: Asymmetry, nonlinearity, and beyond</b> | Submitted to AAAG, currently under review  |
| Co-First Author (2nd) | <b>Spatial Pattern Entropy: A Symbolic Measure for Continuous Spatial Data</b>                                   | Submitted to IJGIS, currently under review |
| First Author          | <b>Unveiling interaction between spatial processes with information decomposition</b>                            | In writing                                 |

|              |                                                                                                      |            |
|--------------|------------------------------------------------------------------------------------------------------|------------|
| First Author | <b>Pattern Causality for Detecting Interactions with the R Package pc</b>                            | In writing |
| First Author | <b>coupling: Coupling Coordination Analysis in R</b>                                                 | Plan       |
| First Author | <b>Spatial causal discovery under latent confounders via deep orthogonal representation learning</b> | Plan       |

Developed Spatial Analysis Toolkit

| Package       | Description                                                           | Language       |
|---------------|-----------------------------------------------------------------------|----------------|
| spEDM         | Spatial Empirical Dynamic Modeling                                    | C++, R         |
| tEDM          | Temporal Empirical Dynamic Modeling                                   | C++, R         |
| infoxtr       | Information-Theoretic Measures for<br>Revealing Variable Interactions | C++, R         |
| pc            | Pattern Causality Analysis                                            | C++, R         |
| gdverse       | Analysis of Spatial Stratified<br>Heterogeneity                       | R, C++, Python |
| sdsfun        | Spatial Data Science Complementary<br>Features                        | R, C++         |
| coupling      | Coupling Coordination Analysis                                        | C++, R         |
| gobi          | General ODE-Based Causal Inference                                    | C++, R         |
| geocomplexity | Mitigate Spatial Bias Through<br>Geographical Complexity              | C++, R, C      |
| HSAR          | Hierarchical Spatial Autoregressive<br>Model                          | C++, R         |
| GD            | Geographical Detectors for Assessing<br>Spatial Factors               | R              |
| geosimilarity | Geographically Optimal Similarity                                     | R              |